Akoizo Original Practice - Form 1 - Answer key

Division B | Disease Detectives | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which measure concerns new cases?

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

A. Incidence

B. Point prevalence

C. Mean disease duration

D. Reporting delay

Correct: A

2. (2 points) RR for the example?

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

A. 4

B. 6

C. 0.3

D. 20

Correct: A

3. (2 points) Which criterion creates circularity when testing a food exposure?

A. Requiring that food consumption to be a case

B. A defined onset interval

C. A standardized symptom threshold

D. A documented event population

Correct: A

4. (2 points) Ten cases in 20 people versus ten in 200 means…

A. Different proportions

B. Equal risk necessarily

C. No denominator is needed

D. The second group has higher risk

Correct: A

5. (2 points) Sensitivity's denominator is…

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

A. All reference-diseased people

B. All test-positive people

C. All reference-negative people

D. Only true positives

Correct: A

6. (2 points) A more sensitive broad definition may…

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

A. Include more unrelated illness

B. Eliminate every false positive

C. Always improve causal inference

D. Guarantee laboratory confirmation

Correct: A

7. (2 points) Reported cases may overrepresent…

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

A. Severe or readily tested illness

B. Every infection equally

C. Only uninfected people

D. A fixed fraction of every population

Correct: A

8. (2 points) OR for the example?

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

A. 6

B. 4

C. 0.4

D. 0.1

Correct: A

9. (2 points) Missing onset date should be…

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

A. Recorded as unknown or pending

B. Invented from the report date

C. Automatically coded no illness

D. Always ignored

Correct: A

10. (4 points) Calculate 10/100 as an outbreak proportion.

answer: 0.10 or 10% over the defined week among initially at-risk people. (4 points).

11. (4 points) Compute the risk difference.

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

answer: 0.4 - 0.1 = 0.3, or 30 percentage points. (4 points).

12. (4 points) Write the example's logic.

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

answer: Attendance AND onset in days 1-3 AND (vomiting OR at least three loose stools). (4 points).

13. (4 points) Why can prevalence rise without increased incidence?

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

answer: Longer disease duration or improved survival can increase the number of existing cases. (4 points).

14. (4 points) Explain why OR differs from RR here.

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

answer: Odds and risks differ, especially with common outcomes; exposed risk 0.4 gives odds 2/3 rather than 0.4. (4 points).

15. (4 points) Why version a changed definition?

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

answer: To reproduce classification and distinguish changes in criteria from changes in disease occurrence. (4 points).

16. (4 points) Distinguish infection and reporting dates.

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

answer: Infection is acquisition of the agent; reporting occurs after detection and administrative processes and can be substantially later. (4 points).

17. (4 points) State RR = 4 in context.

answer: The exposed group had four times the observed illness risk of the unexposed group over the defined follow-up; this alone does not prove cause. (4 points).

18. (4 points) How can two reviewers test clarity?

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

answer: Apply the same definition independently to identical records and resolve disagreements by clarifying operational criteria. (4 points).